

Unish Aryal

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Summary

Software and AI engineer with 3 years of experience building and deploying full-stack systems, from LLM-powered pipelines to production web applications. Resourceful problem-solver who thrives on tackling engineering and system architecture challenges, with a genuine drive to learn and apply new tools quickly.

Education

Caldwell University

B.S. Computer Science | GPA: 3.83 / 4.0

Caldwell, NJ

Graduated May 2026

- **Awards & Honors:** Presidential Scholar, **Dean's Academic List**

- **Relevant Coursework:** Data Structures & Algorithms, Databases, Operating Systems, UX/UI Human-Centered Design, Computer Organization, Probability & Statistics

Experience

HumanityAI

AI Systems Engineer

Upper Montclair, NJ

Jan 2026 – Jun 2026

- Architected a hybrid rule-based + LLM pipeline for categorizing **10,000+** bank transactions, using approximate string matching to route only unseen merchants to the model, reducing LLM inference calls by **90%+** and batch processing time from **5 minutes to under 30 seconds** through continuous batched local inference
- Deployed and self-hosted an **8B-parameter** LLM on an NVIDIA DGX Spark, configuring the GPU-accelerated inference stack and runtime environment to provide low-latency, privacy-preserving inference for sensitive financial data
- Designed a transaction classifier combining MCC rules with direction-aware merchant matching, achieving **99%** classification accuracy on a labeled validation set
- Built a taxonomy validation layer that verifies LLM-generated labels against a fixed category set before caching, preventing hallucinated categories from propagating into future classifications

Wylight Technology

Software Engineer Intern

Mount Airy, NC

Aug 2024 – Dec 2024

- Contributed to a communication platform serving 200+ concurrent users—implemented group video and scheduling features across Python microservices and Node.js APIs, containerized with Docker and deployed on AWS following Agile sprint cycles tracked in Jira and GitHub
- Resolved a critical video performance issue in group calls—engineered an active speaker detection system to dynamically prioritize video rendering and optimized WebRTC parameters for stream clarity, cutting call latency by 90% (800ms → 80ms) and restoring a seamless experience for live users
- Improved cloud-based service reliability through test coverage across 3 microservices (Jest, pytest) and CI/CD-enforced quality gates—catching 15+ regression bugs pre-deployment and raising engineering standards across the team

Selected Projects

Social Pulse — Cloud-Deployed Sentiment Analysis Platform | *Claude code, Node.js, React.js, PostgreSQL, Redis*

- Built a sentiment analysis platform designed for marketing teams to gauge public perception of newly launched products — deployed on AWS EC2 with 5 containerized microservices and an automated data pipeline processing 400+ articles per cycle
- Achieved 100× latency reduction (500ms → 5ms) through Redis caching, PostgreSQL index tuning on complex JOINS/aggregations, and intelligent cache invalidation, enabling near-instant dashboard refreshes
- Built a CI/CD pipeline with GitHub Actions automating test execution (Jest + pytest), TypeScript validation, and zero-touch Docker deployment to AWS EC2, reducing manual release overhead and enforcing code quality on every commit

StuNotes — AI-Powered Full-Stack Web Application | *Python, FastAPI, Flask, TypeScript, React.js, SQLAlchemy, PostgreSQL*

- Led a 6-person team as project manager and backend engineer to build StuNotes, a full-stack application that transcribes lectures into structured study guides and adaptive quizzes, solving a common student productivity challenge
- Reduced LLM token costs by 40% by building an NLTK preprocessing pipeline before Gemini API calls and serving results through versioned FastAPI/SQLAlchemy REST endpoints
- Re-architected the backend service layer mid-project to support stateful adaptive quiz delivery, coordinating frontend and backend development without delaying the 8-week project timeline

Technical Skills

- **Cloud & DevOps:** AWS (EC2, S3, Cloud Practitioner), Kubernetes, Docker, GitHub Actions, CI/CD
- **Databases:** PostgreSQL (query optimization, indexing), MySQL, MongoDB, Redis
- **Engineering Practices:** Agile/Scrum, Git, Jira, unit testing, regression testing (Jest, pytest)

Leadership & Achievements

- **Vice President,** Computer Science Club — Caldwell University
- **Certifications:** AWS Cloud Practitioner, Azure API Management